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9/16/2026

## Week 1 Process Journal

So, I use Google Gemini and sky it “write a c program that calculates and display the area of a rectangle with length 12.5 and width 8.3 then prints the result with an appropriate message”, and it said:

Here is the C program that calculates and prints the area of the rectangle :

```
#include <stdio.h>

int main() {
    // Define the dimensions of the rectangle
    float length = 12.5;
    float width = 8.3;

    // Calculate the area
    float area = length * width;

    // Display the result with an appropriate message
    printf("The area of a rectangle with length %.1f and width %.1f is %.2f.\n", length, width,
area);

    return 0;
}
```

### Output

When you compile and run this program, it will output:

The area of a rectangle with length 12.5 and width 8.3 is 103.75.

And when my partner at that time use chatgpt to ask same question, it use “double” instead if “float”, so I ask Gemini: whats float for, and it response:

In C programming, **float** is a data type used to store numbers that have a decimal point (also known as "floating-point numbers").

Here is why we used it in your program:

- The dimensions you provided were **12.5** and **8.3**.
- If we had used the `int` (integer) data type instead, the computer would have chopped off the decimal parts. It would only see **12** and **8**, which would give you the wrong calculated area (96 instead of 103.75).

and it gives me this chart and the following :

## Quick Comparison

| Data Type    | What it stores        | Examples          | Placeholder in printf |
|--------------|-----------------------|-------------------|-----------------------|
| <b>int</b>   | Whole numbers only    | 5, -42, 1000      | %d or %i              |
| <b>float</b> | Numbers with decimals | 12.5, 3.14, -0.01 | %f                    |

**Formatting Tip:** In the code I wrote for you, I used `%.1f` and `%.2f` inside the `printf` statement. The `.1` and `.2` tell the program exactly how many decimal places to print on the screen so it looks clean!

So I understand that using “float” allows to more accurately calculate and measure the area of a rectangle with length 12.5 and width 8.3, although using double is great, but the data to storage or the number to print are different intern of having a decimal point or not.